**DEPARTMENT OF COMPUTER ENGINEERING**
Course: CPE312 Computer Networks
Deadline: October 25, 2026, 23:59

# Lab 4: Multi-threaded TCP Socket Chat Server

## 1. Assignment Description & Objective

In this assignment, students are required to implement a robust multi-threaded or I/O multiplexed TCP chat server and client in C or Python.

The server must handle at least 5 concurrent connected users without blocking, broadcast messages to all active channels, and support user commands (/list, /private, /quit).

Students must capture TCP packet flows using Wireshark to verify the three-way handshake (SYN, SYN-ACK, ACK) and connection termination (FIN-ACK).

Deliverables include source code, Makefile, and a technical PDF report containing packet analysis diagrams.

## 2. Grading & Evaluation Criteria

• Correct TCP Protocol Implementation & Concurrency (40%)

• Packet Trace Analysis with Wireshark (30%)

• Clean Code Structure & Robust Error Handling (15%)

• Documentation & Lab Report Quality (15%)

*Note: Please deconstruct this assignment into daily actionable tasks using TaskFlow AI to avoid procrastination and burnout.*